(rejected of course). But after three decades of false starts, erroneous readings and lack of support, 1956 witnessed one small victory for the cause – they had a pop-culture term for it now – Cold Fusion.

The Fall of Cold Fusion

Thirty three years later, 1989 saw the first signs of hope as electro-chemists Martin Fleischmann and his student Stanley Pons declared that they had successfully achieved cold fusion. Their experiment involved conducting an electrolysis process using a palladium cathode and deuterium in an insulated vessel designed to measure process heat.

The experiment lasted weeks while current was continuously supplied and the deuterium replenished until suddenly something strange happened. The energy balance of the process shifted and the process heat increased from 30°C to 50°C, with no change in the current supply. The duo quickly realised the commercial potential of their experiment and published their finding in March 1989.

And then, everything went to hell.

The media storm following the Fleischmann-Pons experiment captured the world's imagination. The possibility of a near infinite supply of cheap energy invited interest and scrutiny from all corners of the world - scientific and commercial. But the glory was short lived as scientists across the world were unable to replicate the experiment. In addition to this setback the discovery was burdened by the complete absence of supporting conjectures or theories that could explain how the excess heat was generated.

Even reputed institutions such as CalTech and CERN declaratively stated that the experiments just didn't work. Instances of scientists publishing concurrent findings in support of Fleischmann-Pons and then retracting their support cast doubt on cold fusion as a legitimate science. The chaos surrounding the academic community's divide over the subject led "cold fusion" to become "fusion confusion", and scientists started considering the research to be incompetent and delusional. The final nail in the proverbial coffin was when the pejorative of "pathological science" ultimately spelled doom for the newly blossoming field of science.

By 1991, cold fusion had been firmly relegated to the stature of fringe science along with subjects like martian canals and crystal healing. The scientific community was divided on near religious grounds, as the National Cold Fusion Institute claimed 92 research groups across 10 countries had successfully replicated the experiment but offered no proof to the non-believers.

As the negative reviews and media turned cold fusion into a joke, Fleischmann and Pons left the United States to continue their research for the Toyota Motor Corporation. Over the years they spent USD $44 million without any success. The National Cold Fusion Institute that had been set up in August 1989 also dried up and was shut down within 22 months of operations. For all practical purposes, cold fusion was dead.

How well would you sleep with a fusion reactor in your home if no one but the designer knows how it works? No customer service in a nuclear meltdown. Sorry.

The Prodigal Son Returns, Still Prodigal

For a long time after the Fleischmann-Pons fiasco, obscure groups of scientists would continue their research into the field of cold fusion with no real results. But in 2008 a curious Italian engineer called Andrea Rossi and his E-Cat (Energy Catalyzer) device started gathering interest – commercially and scientifically.

The device was (and sort of is) purported to be the first fully functional and commercially viable cold fusion device. The description of how the device works involves excited hydrogen, a mystery catalyst, a sealed contraption and the ability to let hydrogen atoms "penetrate" nickel and transform it into copper. The energy produced is allegedly self-sustaining and many orders of magnitude larger than the most common fuel sources available.

The miracle device saw significant interest in scientific and commercial circles but given the deep scar from the Fleischmann-Pons affair, community reactions were cautious. Scientists wanted to be sure that whatever Rossi was peddling was legitimate and not another overblown curiosity. Even agencies such as NASA, the U.S. Naval Research Lab and DARPA got involved to verify the claim. But Rossi declined, maintaining a reclusive and secretive attitude over the details of his device, only allowing highly controlled demonstrations that left many questions unanswered.

It was also discovered that Andrea Rossi's engineering degree was from the corrupt and debunked Kensington University in California (shut down in 1996) and the fact that during the 1980s Rossi proclaimed to invent a machine that magically transformed garbage in to oil. That claim resulted in a lawsuit, a "poison factory", 77,000 tonnes of toxic waste and a USD $50 million clean up cost.

As of December 2013 none of that has discouraged the Italian entrepreneur. And